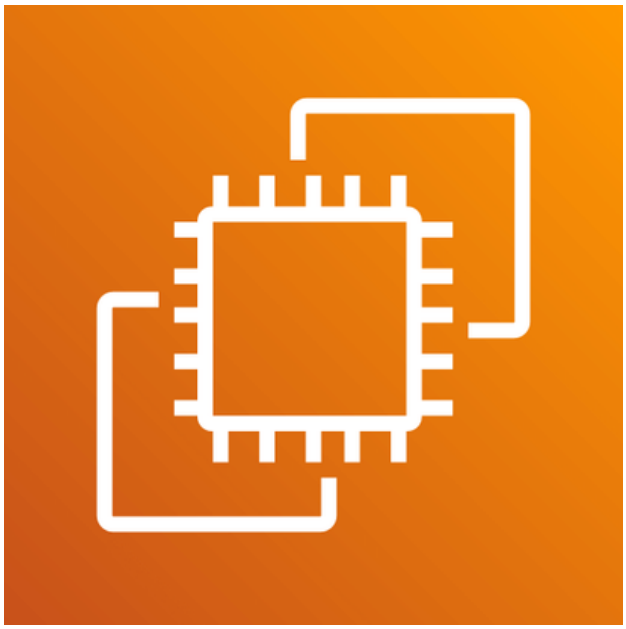


AWS EC2

& Security Groups

Concept Overview:

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About EC2:

Amazon EC2 is a cloud service that provides on-demand, scalable virtual servers.

You can quickly launch, scale up for high demand, and scale down when not needed, saving costs.

Each EC2 instance is a virtual server, and its instance type decides the amount of CPU, memory, storage, and networking it gets.



EC2 Features:

- Instances: Virtual servers.

- Amazon Machine Images (AMIs): Preconfigured templates for your instances that package the components you need for your server (including the operating system and additional software).

- Instance types: Various configurations of CPU, memory, storage, networking capacity, and graphics hardware for your instances.

- Amazon EBS volumes: Persistent storage volumes for your data using Amazon Elastic Block Store (Amazon EBS).

- Instance store volumes: Storage volumes for temporary data that is deleted when you stop, hibernate, or terminate your instance.

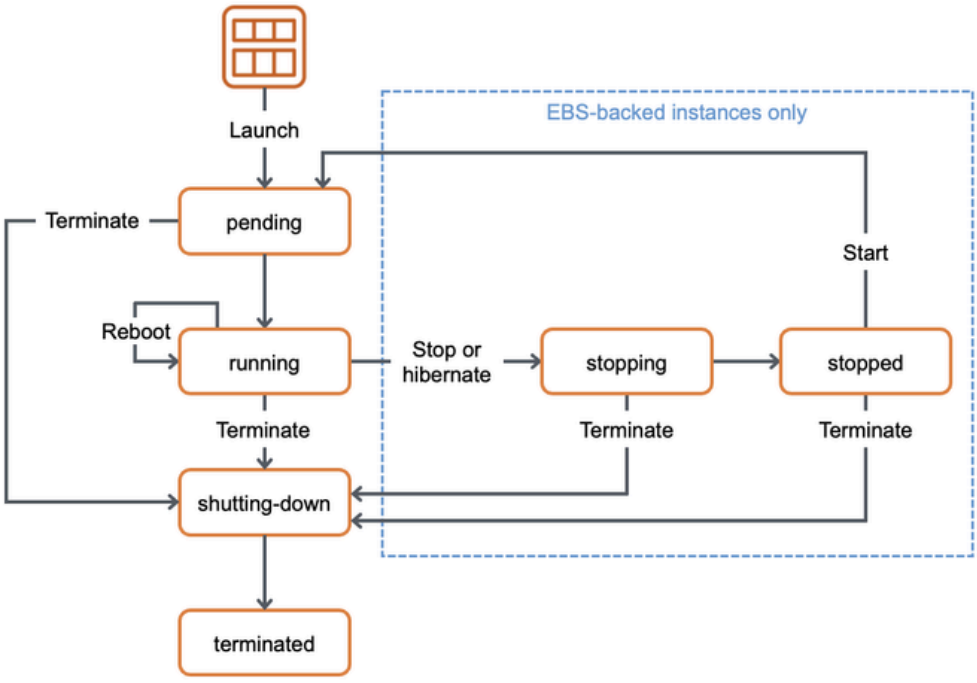
- Key pairs: Secure login information for your instances. AWS stores the public key and you store the private key in a secure place.

- Security groups: A virtual firewall that allows you to specify the protocols, ports, and source IP ranges that can reach your instances, and the destination IP ranges to which your instances can connect.

EC2 Lifecycle:

- Pending: The instance is being provisioned and prepared for launch.
- Running: The instance is operational and actively serving your applications.
- Stopping: The instance is undergoing a graceful shutdown process.
- Stopped: The instance is powered off, but its resources remain allocated.
- Shutting down: The instance is preparing to be terminated.
- Terminated: The instance is permanently deleted, and its resources are released.

For better understanding, follow the diagram:



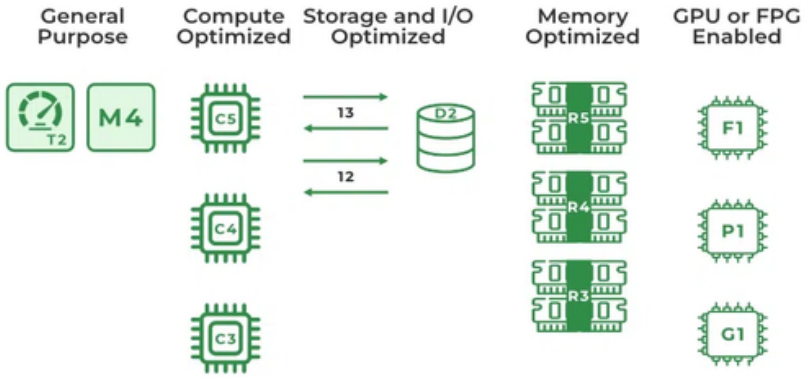
EC2 Instance Types:

The AWS EC2 Instance Types are as follows:

1. General Purpose Instances
2. Compute Optimized Instances
3. Memory-Optimized Instances
4. Storage Optimized Instances
5. Accelerated Computing Instances

Most common used instance type:

- General Purpose (T3, T4g, M5, M6): Most popular for balanced workloads.
- Compute Optimized (C5, C6): Popular for CPU-intensive tasks.
- Memory Optimized (R5, R6): Popular for databases & big in-memory apps.



For more details about EC2 instance types, you can follow the AWS official docs.

https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/instance-types.html

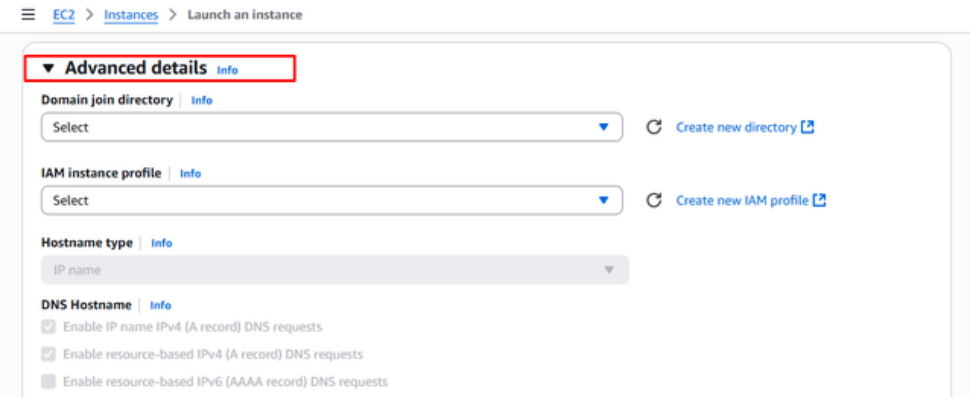
User Data:

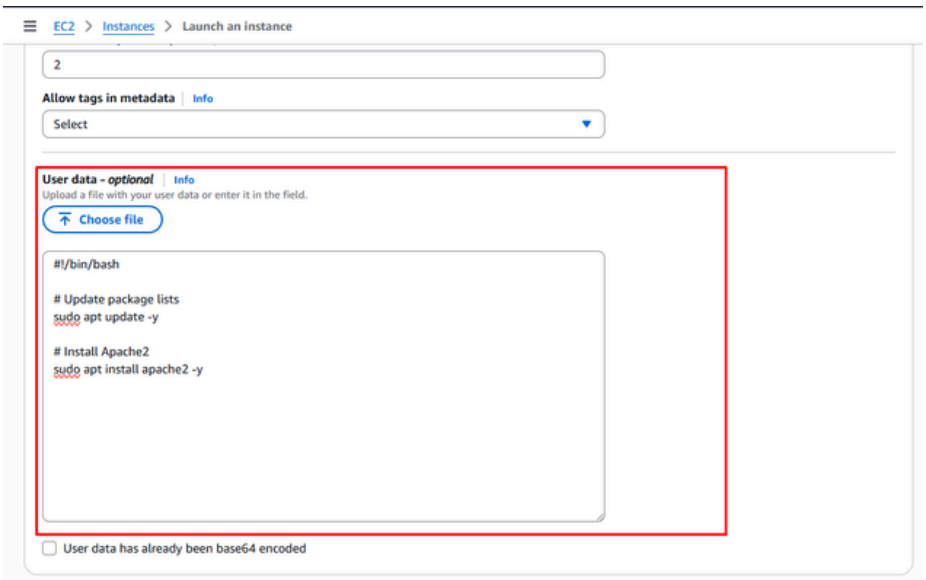
AWS user data is a feature that allows the execution of scripts or commands on an Amazon EC2 instance when it is launched for the first time.

- This script is executed with a user with root privileges
- You can install updates, packages, change configurations and anything you can do with scripts can be done.

User Data helps you to Automation of your tasks.

You find it in Advance Details tabs when you create a new instance.



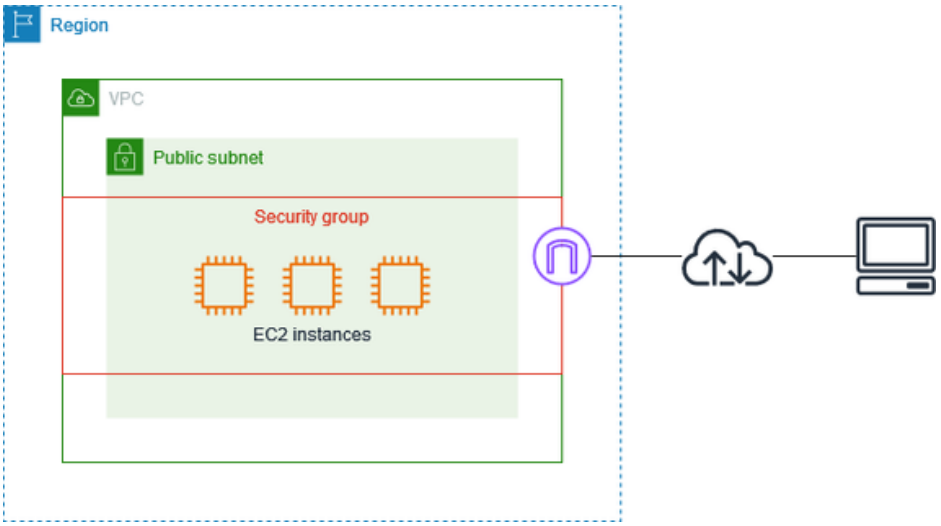


Security Groups:

A security group acts as a virtual firewall for your EC2 instances to control incoming and outgoing traffic.

Inbound rules control the incoming traffic to your instance, and outbound rules control the outgoing traffic from your instance. When you launch an instance, you can specify one or more security groups.

If you don't specify a security group, Amazon EC2 uses the default security group for the VPC. After you launch an instance, you can change its security groups.

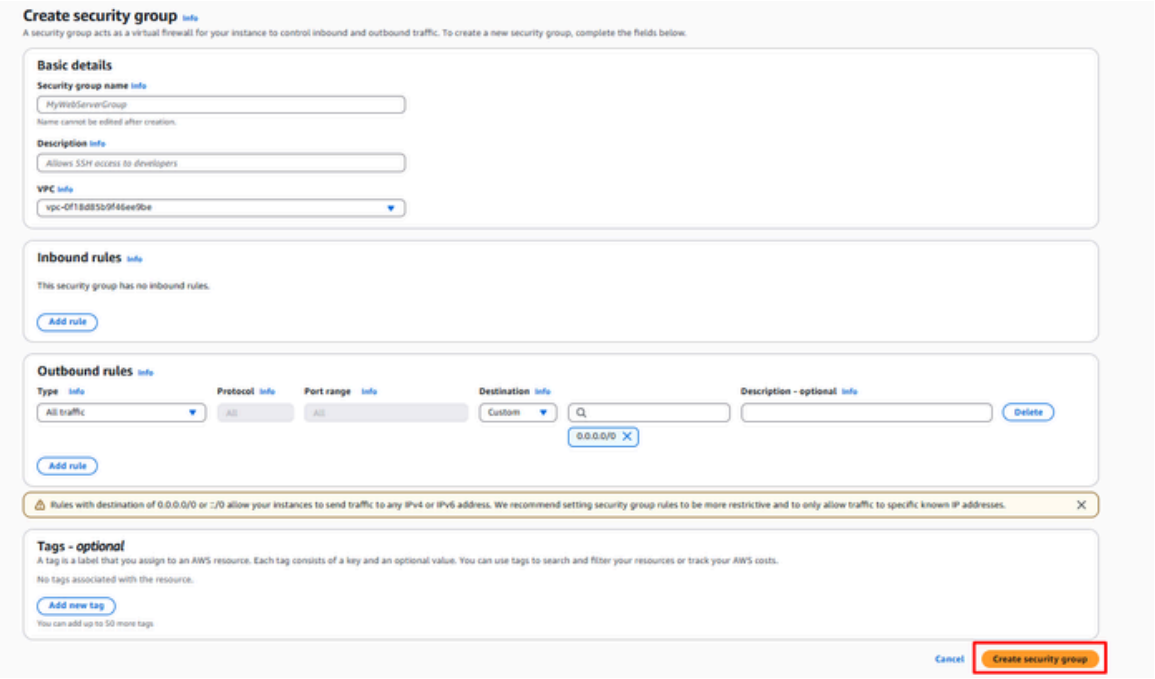


Security Groups:

For create a security group you need to go security group section from EC2 dashboard.

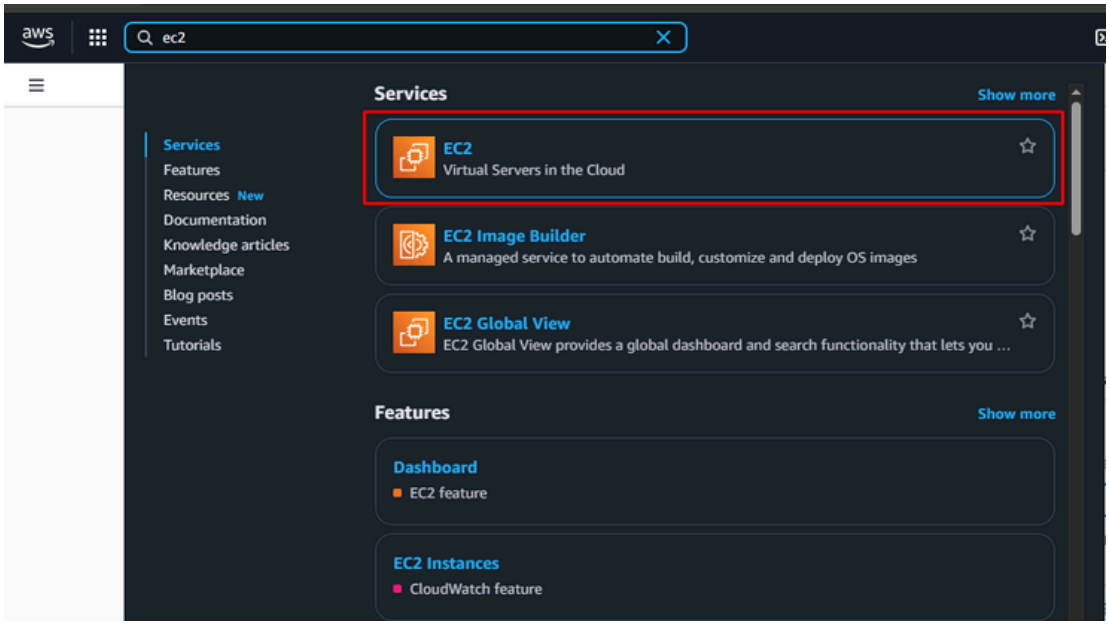


Fill up the input fields based on your needs and click on Create Security Group.

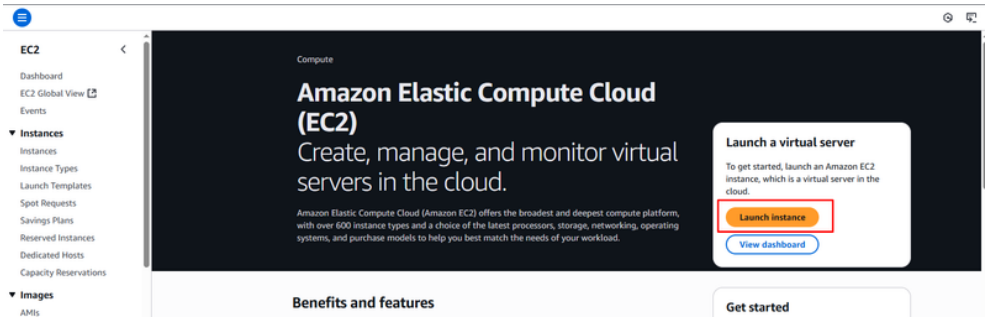


Launch First EC2 Instance:

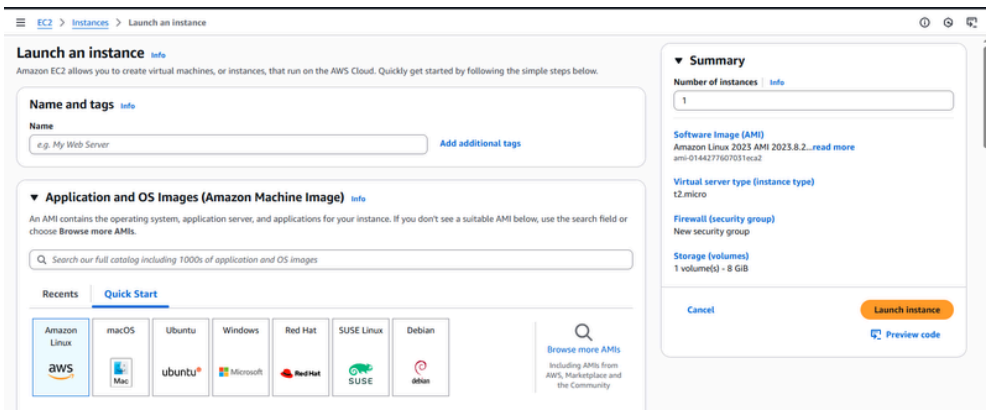
Step: 1 - Log in into your AWS account and search for EC2 service.



Step: 2 - Click on EC2



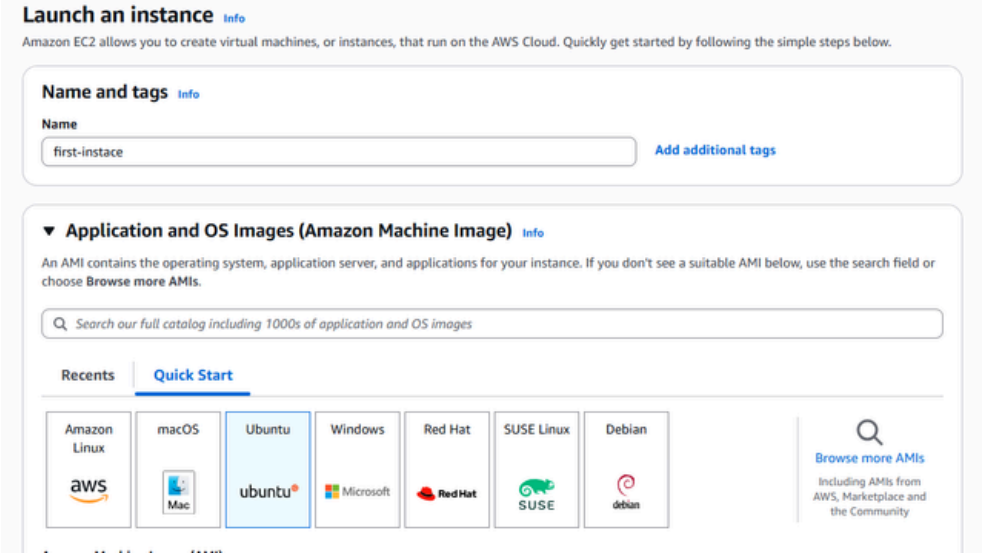
Step: 3 - Click on Launch Instance and get this page



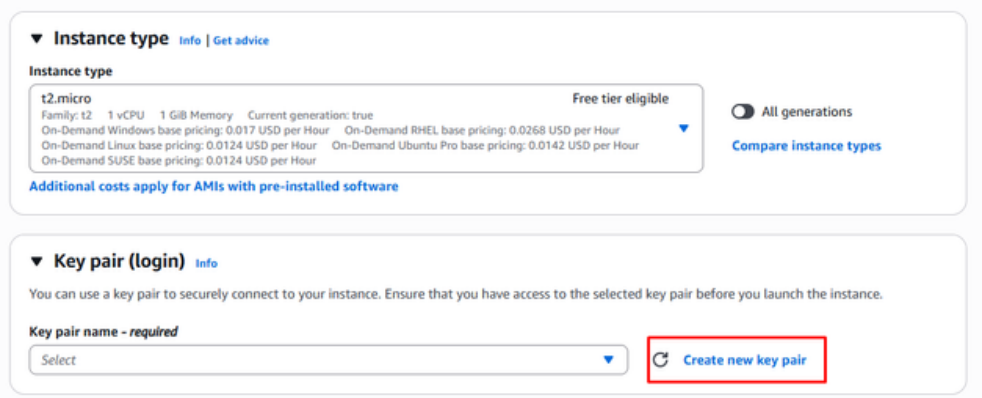
Launch First EC2 Instance:

Step: 4 - Now set configuration for EC2 instance

In my case, i used free tier for launch instance and create new key-pair file for connect with ssh.



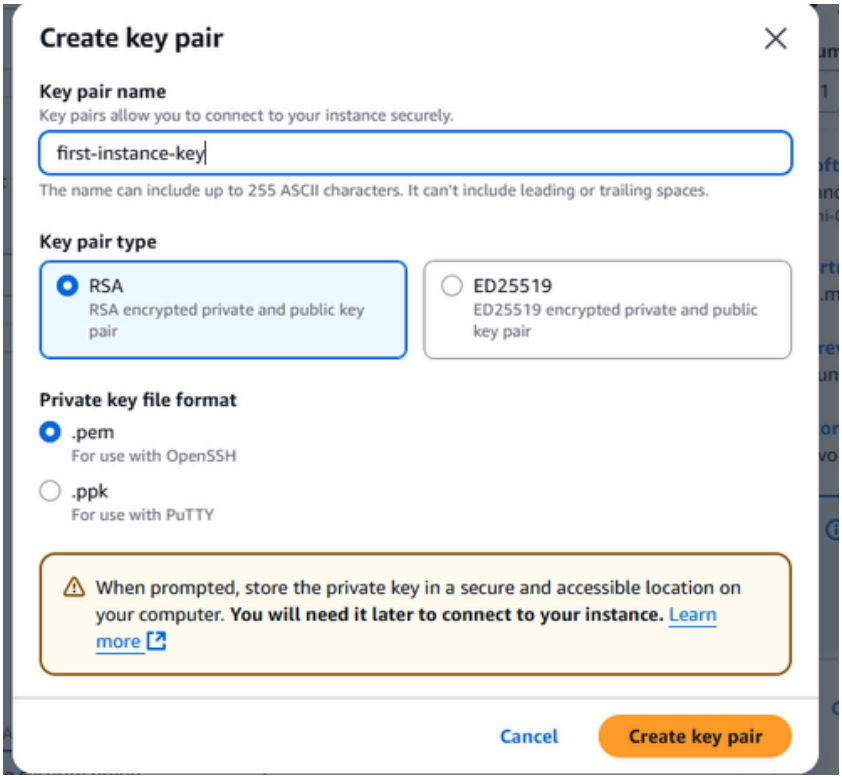
Give the name of your instance, and choose OS. I am choosing ubuntu.



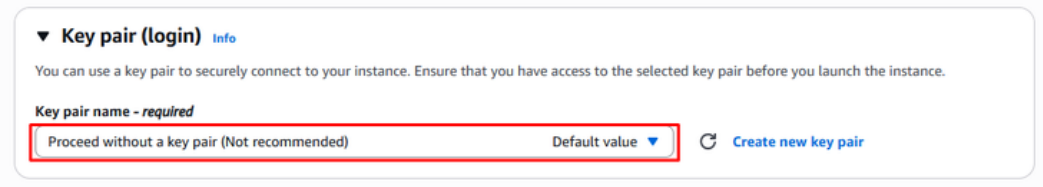
I keep all things as it is. Just create a new key pair for connect to SSH. If you need any other custom configuration you can.

Launch First EC2 Instance:

Name your key pair if you want to create it. For an existing key pair no need naming just select previous one.

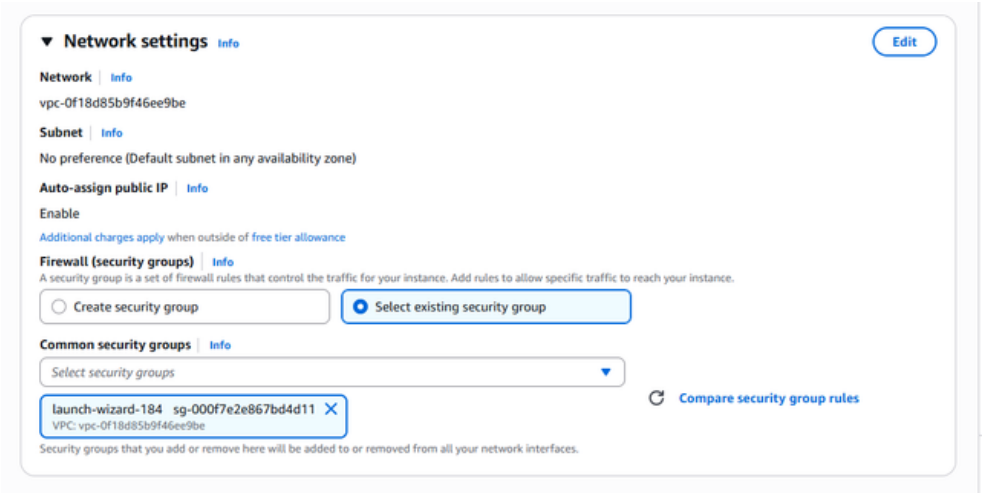


You can launch instance without key pair also.

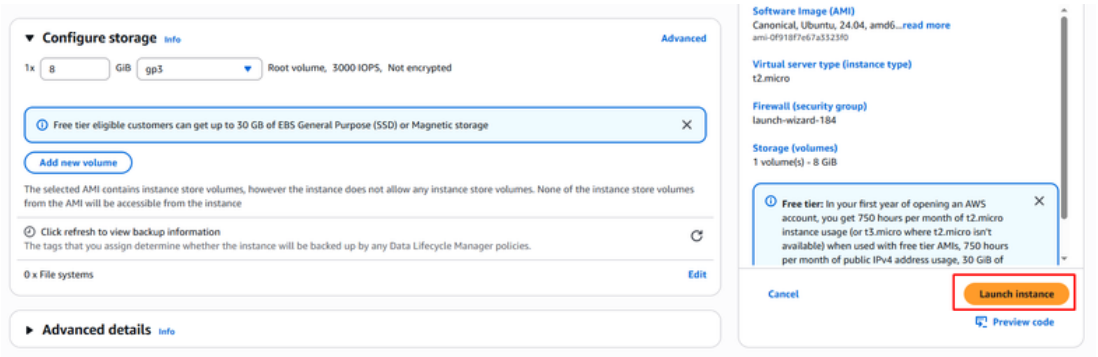


Launch First EC2 Instance:

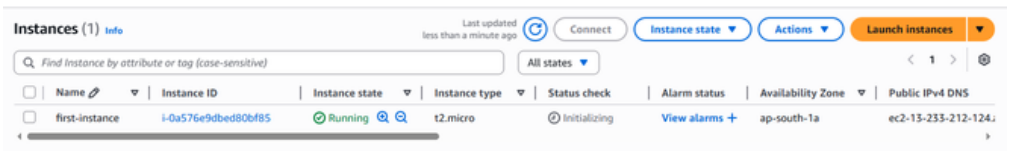
I use an existing security group. If you need, you can creeate new one.



After configure all of things, then click on Lanuch instance button and create it.

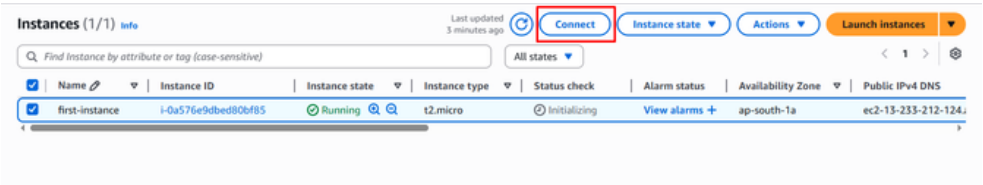


Successfully launch your instance you can see it in your EC2 instance page.

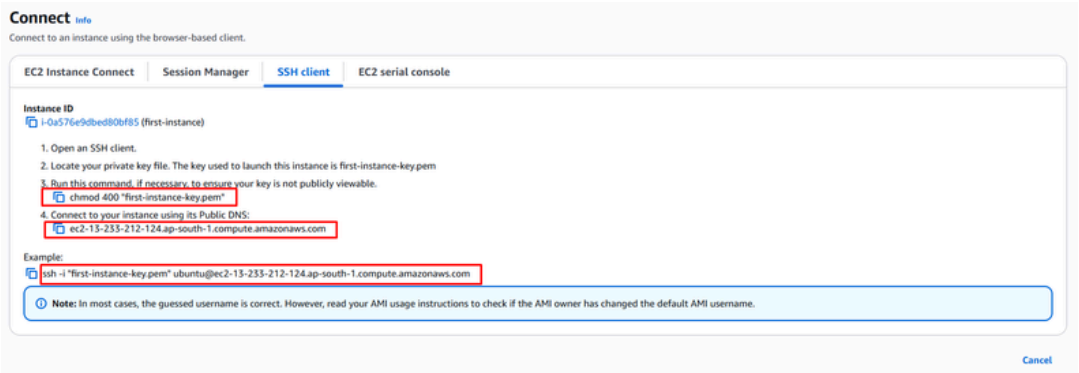


Connect EC2 with SSH:

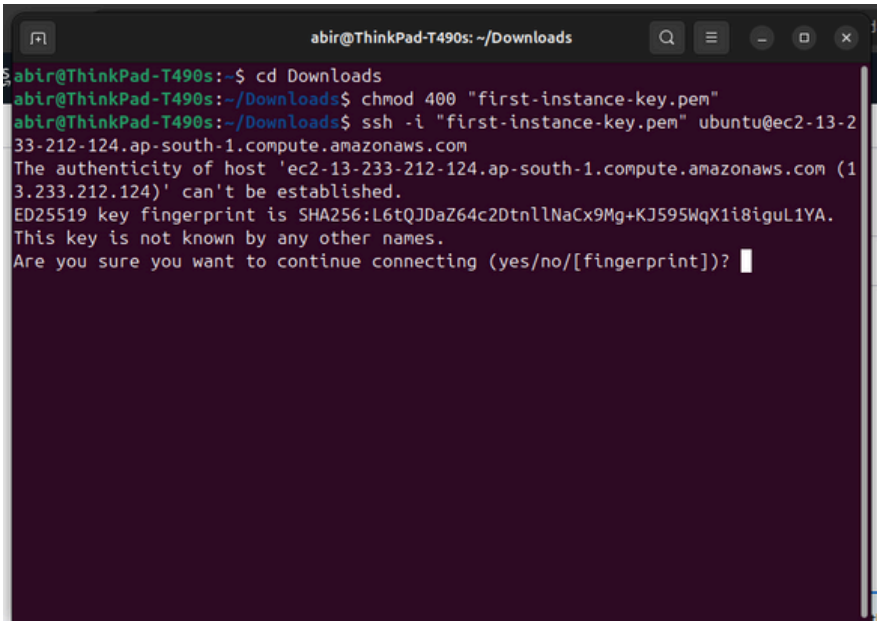
For connect your instance with SSH. First you need to connect your instance.



After connecting your instance, you need to go SSH client tab. From here, you need three info for connecting with ssh, which i marked.



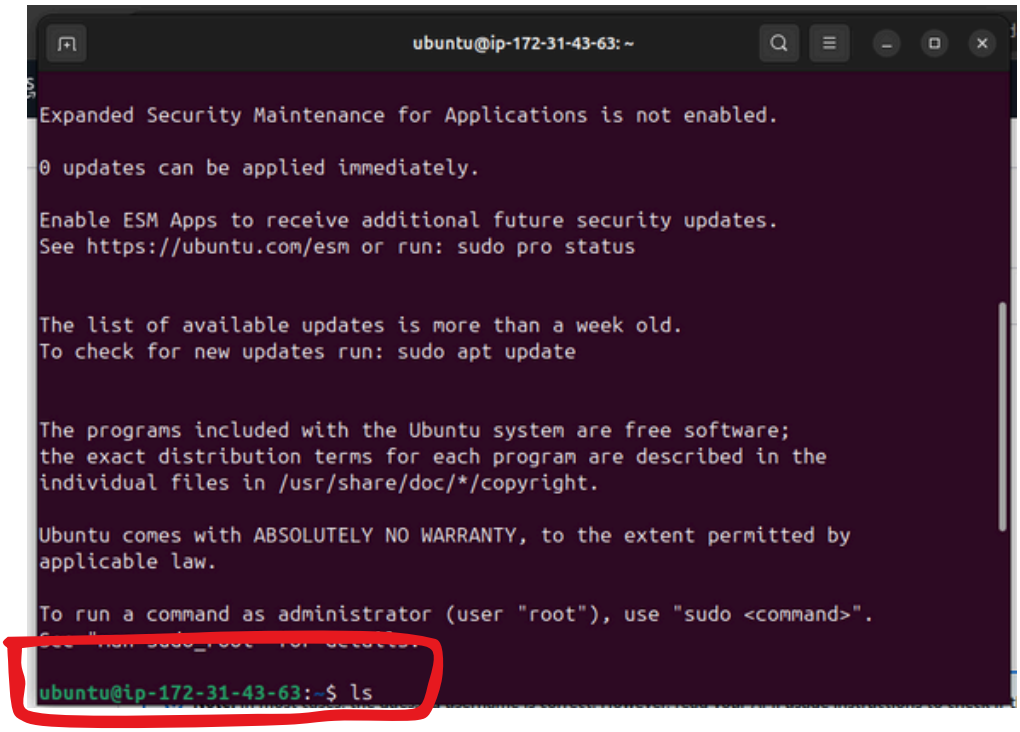
Now open you local machine terminal and connection the EC2 instance.



type yes and press enter to connect.

Connect EC2 with SSH:

Successfully conntect EC2 with SSH.

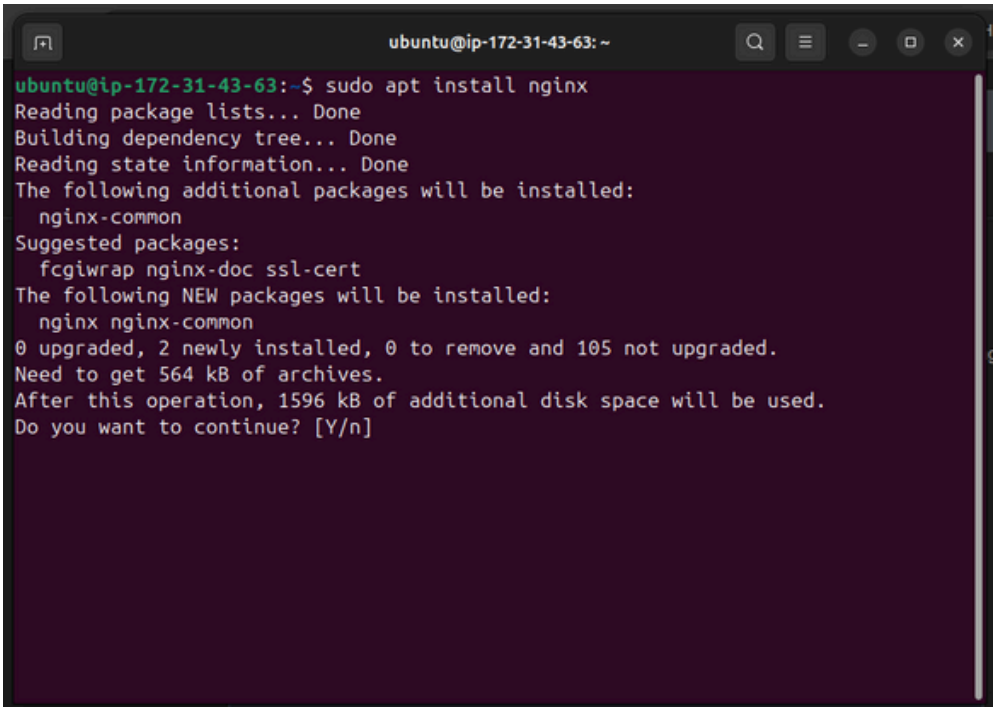


Now it’s time to configure nginx in EC2 instance.

Configure nginx:

For configure nginx in your EC2 you need to run two one for system updates and another for install nginx.

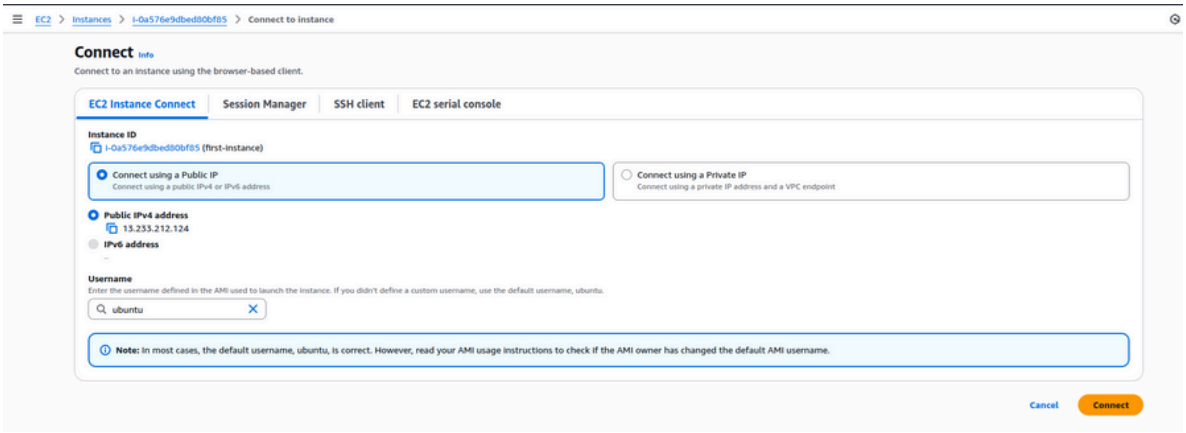
1. sudo apt update
2. sudo install nginx



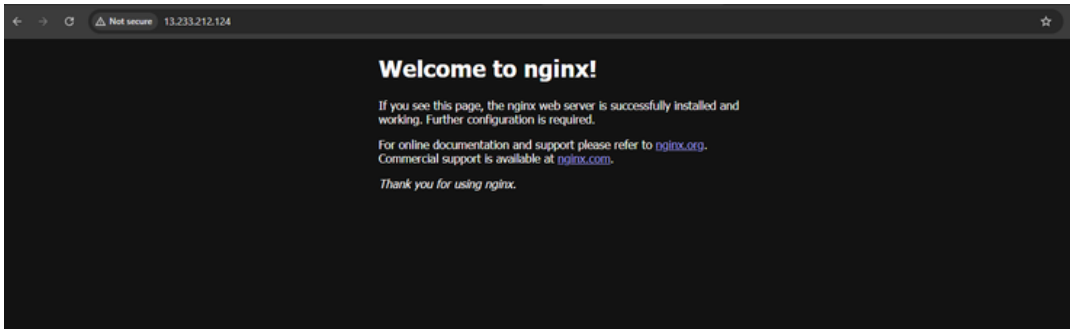
Type y and press enter for full configuration.

Configure nginx:

It’s time to connect EC2 instance and testing.



Successfully configure the nginx.



Thank You

Stay Connect:


/in/alamgirweb11


/alamgirweb11